

# Reds Transactions Time Series

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## Time Series Data

### Time Series Data on Cincinatti Reds Transactions from 2016-2024

```
## Registered S3 method overwritten by 'quantmod':
##   method           from
##   as.zoo.data.frame zoo

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## — Attaching packages ————— fpp2
2.5 —

## ✓ ggplot2    3.5.1    ✓ expsmooth 2.3
## ✓ fma        2.5

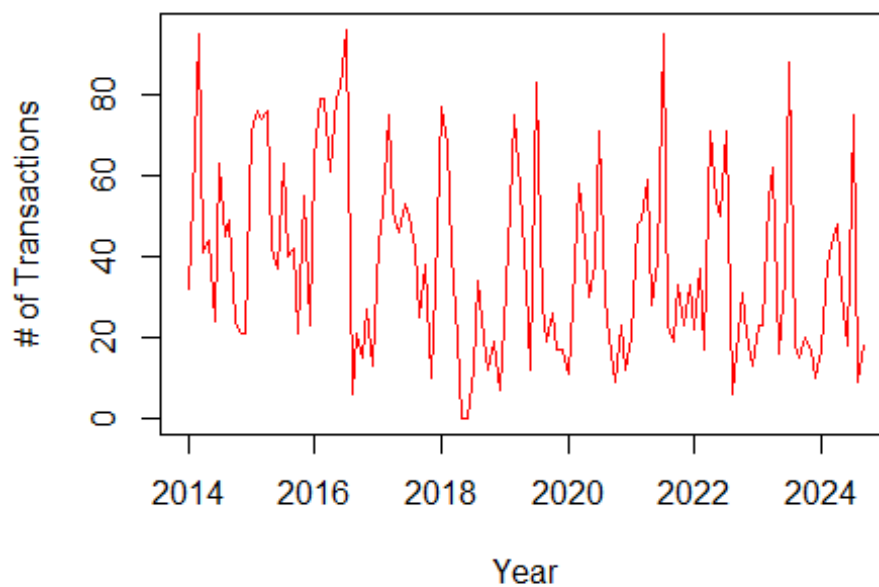
##

## Rows: 129 Columns: 2
## — Column specification
##
## Delimiter: ","
## chr (1): month
## dbl (1): # of transactions
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

## Time Series Data

- My data is tracking the number of transactions the Reds make each month on their 40-man roster.
  - signing
  - activating
  - trading
  - placing on IL
  - optioning
  - etc.
- There are a couple variables that could play a role in impacting the results of how many transactions there are each month:
  - the trade deadlines
  - the pre-season transactions
  - season injuries
  - etc.

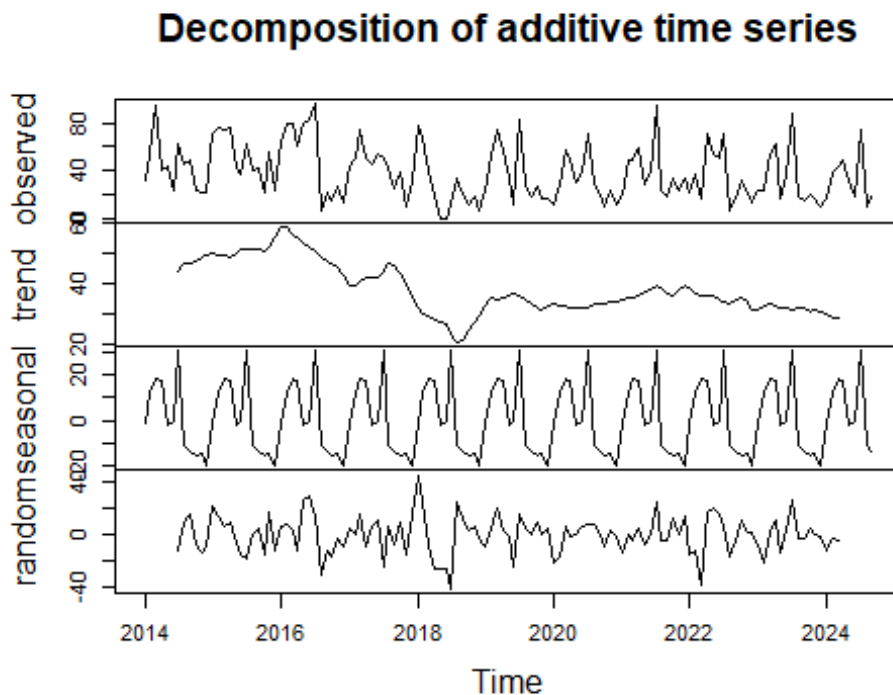
### Cincinnati Reds Transactions From 2014-2024



The transaction time series displayed clear yearly seasonality with recurring spikes and dips throughout each MLB season. These recurring patterns likely reflect predictable baseball calendar events such as preseason roster activity, trade deadline transactions, injury-related roster movement, and postseason organizational adjustments.

Although some mild fluctuations in overall transaction frequency were observed across years, the series remained relatively stable without strong long-term upward or downward trend behavior.

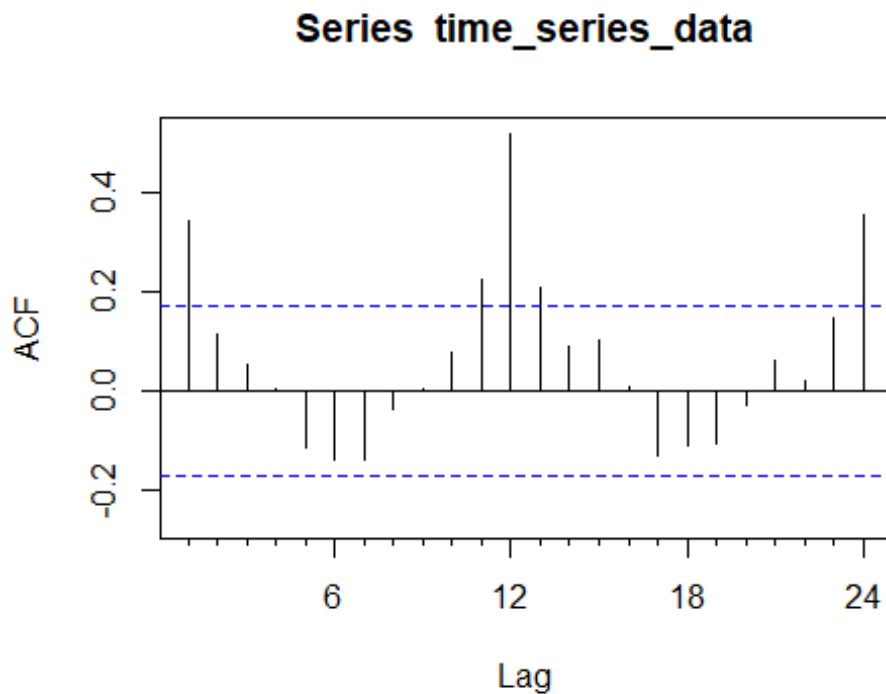
## Decomposition Model



The decomposition model identified both a seasonal and trend component within the transaction data. The trend component suggested a slight long-term decrease in transaction frequency over time, while the seasonal component revealed highly consistent yearly transaction cycles across the dataset.

The strength and consistency of the seasonal component indicated that seasonality plays a major role in forecasting transaction activity and should therefore be incorporated into predictive modeling approaches.

## Auto Correlation Function



The autocorrelation analysis revealed significant seasonal structure within the transaction series. Autocorrelation values exceeded the confidence bands at lag 12, confirming strong yearly seasonality within the data. Additional autocorrelation at lag 24 suggested the possibility of recurring multi-year transaction behavior patterns.

These findings reinforced the importance of incorporating seasonal terms into forecasting models rather than relying solely on trend-based approaches.

## Augmented Dickey-Fuller Test

```
## Warning in adf.test(time_series_data): p-value smaller than printed p-  
value  
  
##  
## Augmented Dickey-Fuller Test  
##  
## data: time_series_data  
## Dickey-Fuller = -5.7796, Lag order = 5, p-value = 0.01  
## alternative hypothesis: stationary
```

An Augmented Dickey-Fuller (ADF) test was conducted to evaluate stationarity within the transaction series. The test produced a p-value of 0.01, allowing rejection of the null

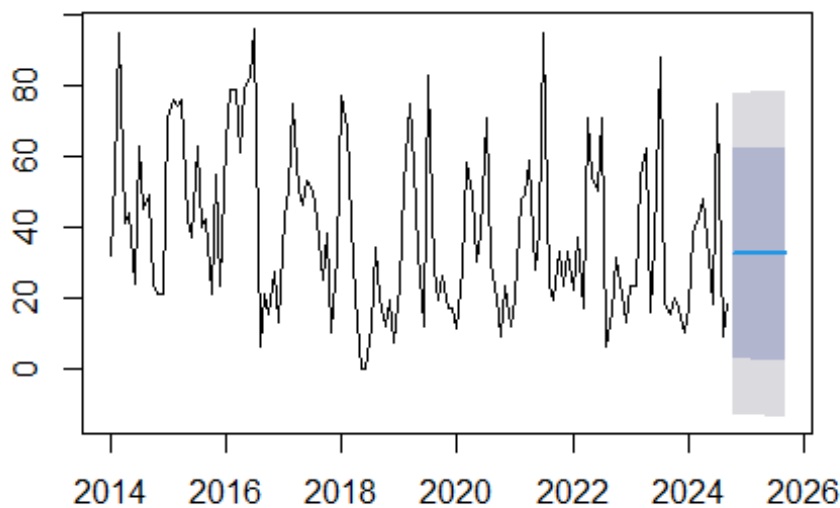
hypothesis of non-stationarity. These results suggest that the series remains statistically stationary over time, with relatively stable mean behavior and limited variance changes across seasons.

The stationarity of the data supported the use of ARIMA-based forecasting approaches without requiring extensive differencing adjustments.

## Exponential Smoothing

### SES (Simple Exponential Smoothing) Model

Forecasts from Simple exponential smoothing



| ## | Point    | Forecast | Lo 80    | Hi 80    | Lo 95     | Hi 95    |
|----|----------|----------|----------|----------|-----------|----------|
| ## | Oct 2024 | 32.27688 | 2.592344 | 61.96143 | -13.12170 | 77.67547 |
| ## | Nov 2024 | 32.27688 | 2.547197 | 62.00657 | -13.19075 | 77.74452 |
| ## | Dec 2024 | 32.27688 | 2.502119 | 62.05165 | -13.25969 | 77.81346 |
| ## | Jan 2025 | 32.27688 | 2.457108 | 62.09666 | -13.32853 | 77.88230 |
| ## | Feb 2025 | 32.27688 | 2.412166 | 62.14160 | -13.39726 | 77.95103 |
| ## | Mar 2025 | 32.27688 | 2.367290 | 62.18648 | -13.46589 | 78.01966 |
| ## | Apr 2025 | 32.27688 | 2.322483 | 62.23129 | -13.53442 | 78.08819 |
| ## | May 2025 | 32.27688 | 2.277742 | 62.27603 | -13.60285 | 78.15662 |
| ## | Jun 2025 | 32.27688 | 2.233067 | 62.32070 | -13.67117 | 78.22494 |
| ## | Jul 2025 | 32.27688 | 2.188459 | 62.36531 | -13.73939 | 78.29316 |
| ## | Aug 2025 | 32.27688 | 2.143917 | 62.40985 | -13.80751 | 78.36128 |
| ## | Sep 2025 | 32.27688 | 2.099441 | 62.45433 | -13.87553 | 78.42930 |

$$S_t = \alpha Y_t + (1 - \alpha) S_{t-1}$$

$$\alpha = .0522$$

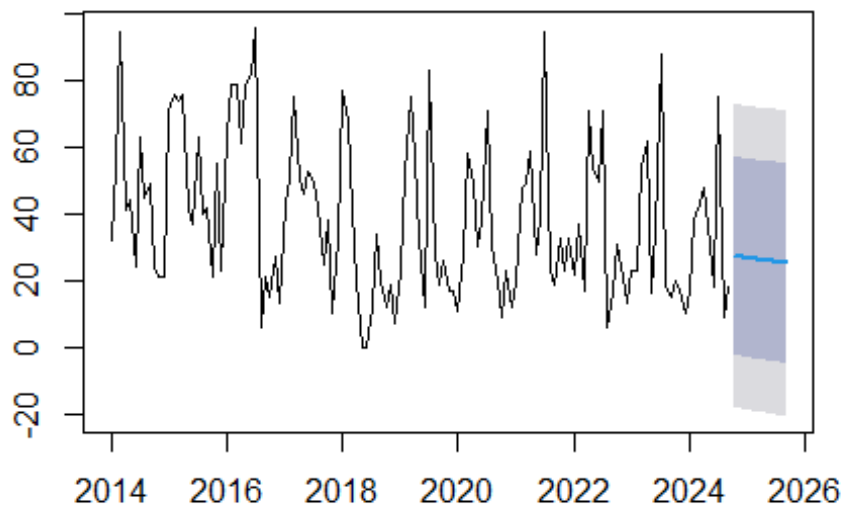
$$S_t = .0552Y_t + .9448S_{t-1}$$

The Simple Exponential Smoothing model did not adequately capture the structure of the transaction data because the method does not account for either trend or seasonality. As a result, the model produced forecasts that failed to reflect the recurring yearly transaction patterns observed throughout the historical data.

Because seasonality represented a major component of the series, SES proved insufficient as a standalone forecasting approach for this dataset.

## Holt's Linear Trend Method

**Forecasts from Holt's method**



| ## | Point    | Forecast | Lo 80     | Hi 80    | Lo 95     | Hi 95    |
|----|----------|----------|-----------|----------|-----------|----------|
| ## | Oct 2024 | 27.41772 | -2.382568 | 57.21801 | -18.15789 | 72.99334 |
| ## | Nov 2024 | 27.21071 | -2.607252 | 57.02867 | -18.39193 | 72.81334 |
| ## | Dec 2024 | 27.00369 | -2.832028 | 56.83942 | -18.62611 | 72.63349 |
| ## | Jan 2025 | 26.79668 | -3.056897 | 56.65026 | -18.86043 | 72.45378 |
| ## | Feb 2025 | 26.58967 | -3.281857 | 56.46119 | -19.09489 | 72.27422 |
| ## | Mar 2025 | 26.38265 | -3.506911 | 56.27221 | -19.32949 | 72.09479 |
| ## | Apr 2025 | 26.17564 | -3.732057 | 56.08333 | -19.56423 | 71.91551 |
| ## | May 2025 | 25.96862 | -3.957296 | 55.89454 | -19.79912 | 71.73636 |

|             |          |           |          |           |          |
|-------------|----------|-----------|----------|-----------|----------|
| ## Jun 2025 | 25.76161 | -4.182628 | 55.70584 | -20.03415 | 71.55736 |
| ## Jul 2025 | 25.55459 | -4.408053 | 55.51724 | -20.26932 | 71.37851 |
| ## Aug 2025 | 25.34758 | -4.633571 | 55.32873 | -20.50463 | 71.19979 |
| ## Sep 2025 | 25.14056 | -4.859183 | 55.14031 | -20.74009 | 71.02122 |

$$\begin{aligned}\hat{Y}_t &= l_{t-1} + b_{t-1} \\ l_t &= \alpha Y_t + (1 - \alpha)(l_{t-1} + b_{t-1}) \\ b_t &= \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}\end{aligned}$$

$$\begin{aligned}\alpha &= .0343 \\ \beta &= .0001 \\ l_{t-1} &= 57.8137 \\ b_{t-1} &= -.1995\end{aligned}$$

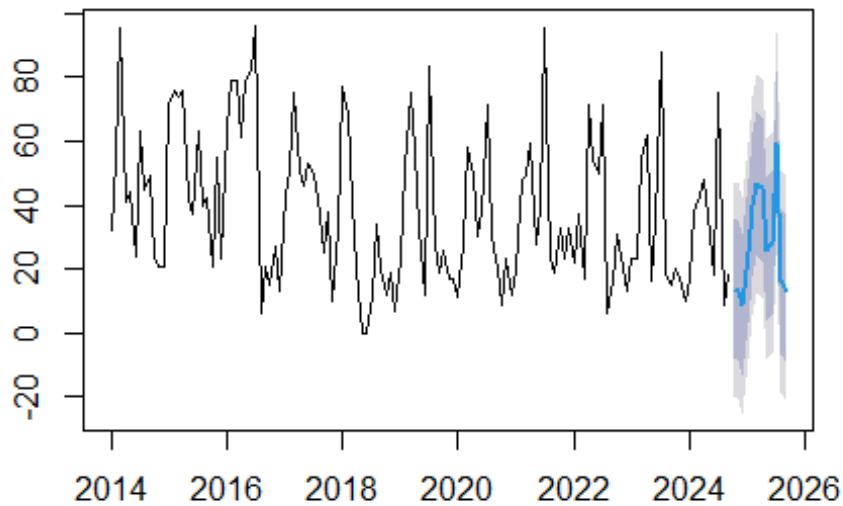
$$\begin{aligned}\hat{Y}_t &= 57.6142 \\ l_t &= .0343(Y_t) + 55.6380 \\ b_t &= .0001(l_t) - .2053\end{aligned}$$

Holt's Linear Trend Method improved upon SES by incorporating a trend component into the forecasting process. The model successfully captured the slight decreasing trend identified in the decomposition analysis; however, it still failed to account for the strong yearly seasonality present within the transaction data.

Although Holt's method produced forecasts more aligned with the long-term trend behavior of the series, the absence of seasonal modeling limited its forecasting performance.

## Holt-Winters (Triple Exponential Smoothing)

### Forecasts from Holt-Winters' additive method



| ##          | Point Forecast | Lo 80      | Hi 80    | Lo 95      | Hi 95    |
|-------------|----------------|------------|----------|------------|----------|
| ## Oct 2024 | 13.326634      | -8.628253  | 35.28152 | -20.250468 | 46.90374 |
| ## Nov 2024 | 13.436938      | -8.597156  | 35.47103 | -20.261302 | 47.13518 |
| ## Dec 2024 | 8.491548       | -13.621673 | 30.60477 | -25.327706 | 42.31080 |
| ## Jan 2025 | 25.417877      | 3.225609   | 47.61014 | -8.522268  | 59.35802 |
| ## Feb 2025 | 39.947481      | 17.676245  | 62.21872 | 5.886566   | 74.00840 |
| ## Mar 2025 | 46.835983      | 24.485858  | 69.18611 | 12.654416  | 81.01755 |
| ## Apr 2025 | 45.300763      | 22.871825  | 67.72970 | 10.998663  | 79.60286 |
| ## May 2025 | 25.881843      | 3.374168   | 48.38952 | -8.540675  | 60.30436 |
| ## Jun 2025 | 28.137344      | 5.551007   | 50.72368 | -6.405478  | 62.68017 |
| ## Jul 2025 | 59.377946      | 36.713020  | 82.04287 | 24.714933  | 94.04096 |
| ## Aug 2025 | 16.250544      | -6.492897  | 38.99399 | -18.532548 | 51.03364 |
| ## Sep 2025 | 13.417652      | -9.404233  | 36.23954 | -21.485410 | 48.32071 |

$$\hat{Y}_t = (l_{t-1} + b_{t-1})S_{t-p}$$

$$l_t = \alpha(Y_t/s_{t-p}) + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$s_t = \delta(Y_t/l_t) + (1 - \delta)s_{t-p}$$

$$\alpha = .0849$$

$$\beta = .0001$$

$$\delta = .0001$$

$$\begin{aligned}\hat{Y}_t &= (l_{t-1} + b_{t-1})S_{t-p} \\ l_t &= .0849 \frac{Y_t}{s_{t-p}} + (.9151)(l_{t-1} + b_{t-1}) \\ b_t &= .0001*(l_t - l_{t-1}) + (.9999)b_{t-1} \\ s_t &= .0001 \frac{Y_t}{(l_{t-1} + b_{t-1})} + (.9999)s_{t-p}\end{aligned}$$

The Holt-Winters Triple Exponential Smoothing model provided substantially improved forecasting performance by incorporating level, trend, and seasonal components simultaneously. Unlike SES and Holt's Linear Trend Method, Holt-Winters successfully reproduced the cyclical transaction patterns observed throughout the MLB calendar.

Forecasted transaction values maintained both the yearly seasonal spikes and the overall long-term trend structure, making Holt-Winters one of the strongest forecasting approaches evaluated within the analysis.

## Model Accuracy

SES Model Accuracy:

| ##              | ME        | RMSE     | MAE      | MPE  | MAPE | MASE     | ACF1      |
|-----------------|-----------|----------|----------|------|------|----------|-----------|
| ## Training set | -2.184319 | 22.98271 | 19.27862 | -Inf | Inf  | 1.127236 | 0.2780426 |

Holt Model Accuracy

| ##              | ME         | RMSE     | MAE      | MPE  | MAPE | MASE     | ACF1     |
|-----------------|------------|----------|----------|------|------|----------|----------|
| ## Training set | -0.5805747 | 22.88994 | 18.79839 | -Inf | Inf  | 1.099156 | 0.289916 |

Holt-Winters Model Accuracy

| ##              | ME        | RMSE     | MAE     | MPE  | MAPE | MASE      | ACF1     |
|-----------------|-----------|----------|---------|------|------|-----------|----------|
| ## Training set | 0.1199806 | 16.03391 | 12.1033 | -Inf | Inf  | 0.7076893 | 0.221398 |

- RMSE: Holt Winters model takes into consideration the large error variables in the prediction process therefore predictions could be off by 16 transactions a month
  - For Simple Exponential Smoothing and Holts Linear Model the predictions are off by 23 transactions a month.
- MAE: Holt Winters model's predictions are off by 12 transactions a month
  - Simple Exponential Smoothing and Holts Linear Model the predictions are off by 19 transactions a month 7 more than the Holt Winters model.
- MASE: Holt Winters model is more accurate than a naive model
  - Simple Exponential Smoothing and Holts Linear Model the value for MASE surpasses 1 meaning that naive model would be more accurate
- ACF1: Holt Winters model has the least correlation in errors

- However Simple Exponential Smoothing and Holts Linear Model correlation in errors is relitevely the same as the Holt Winters

## ARIMA

### ARIMA Estimation

The auto.arima() procedure selected an ARIMA(1,0,0)(0,1,1)[12] model with drift, incorporating both seasonal differencing and seasonal moving-average behavior.

### ARIMA Results

```
## Series: time_series_data
## ARIMA(1,0,0)(0,1,1)[12] with drift
##
## Coefficients:
##          ar1      sma1      drift
##          0.3059 -0.7290 -0.1511
## s.e.  0.0890   0.0855   0.0728
##
## sigma^2 = 298.5: log likelihood = -502.46
## AIC=1012.92  AICc=1013.28  BIC=1023.97
##
## Training set error measures:
##              ME      RMSE      MAE  MPE  MAPE      MASE      ACF1
## Training set 0.3117944 16.24117 11.68952 -Inf  Inf  0.683495 -0.004674693
```

### Mathematical Formula for ARIMA w/ drift:

$$Y_t = \mu + \phi_1 Y_{t-1} + \theta_1 (Y_{t-12} - Y_{t-13}) + \epsilon_t$$

$$\mu = -.1511$$

$$\phi_1 = .3059$$

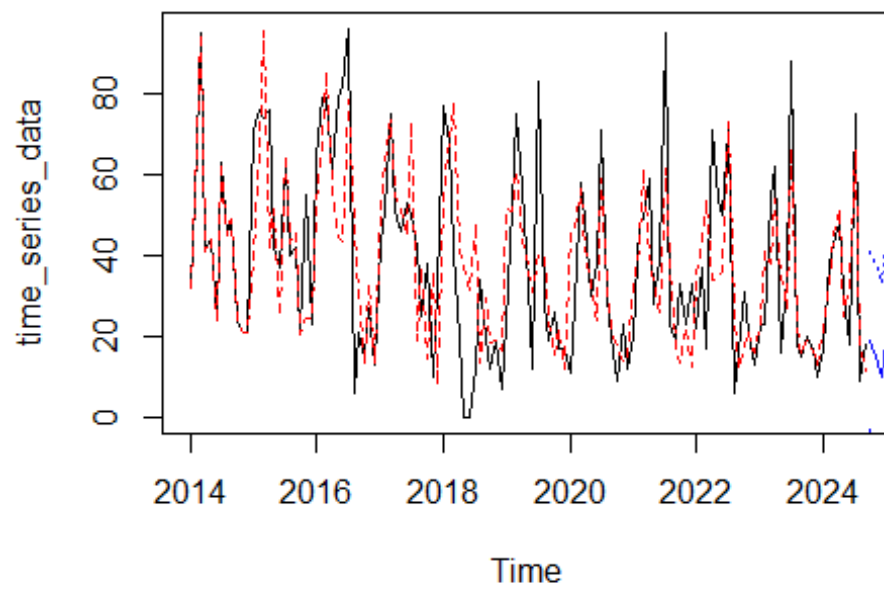
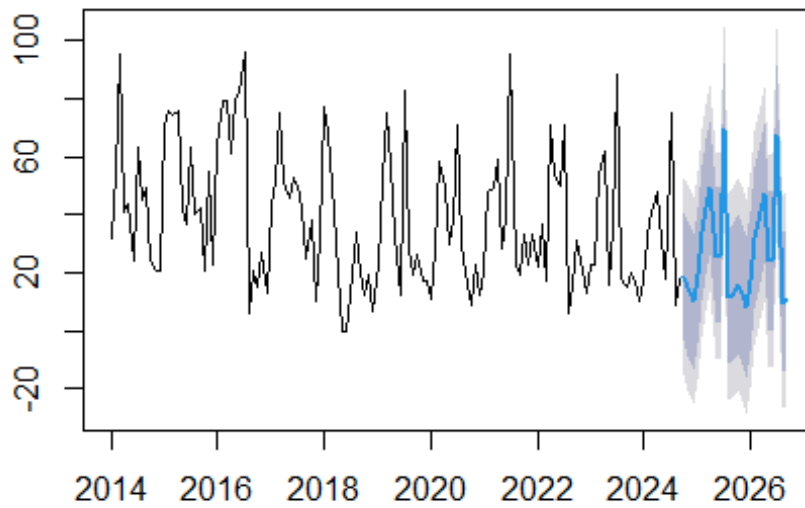
$$\theta_1 = -.7290$$

$$Y_t = -.1511 + 0.3059(Y_{t-1}) - .7290(Y_{t-12} - Y_{t-13}) + \epsilon_t$$

The selected model effectively captured the yearly seasonality, slight trend behavior, and short-term autocorrelation structure present within the transaction series.

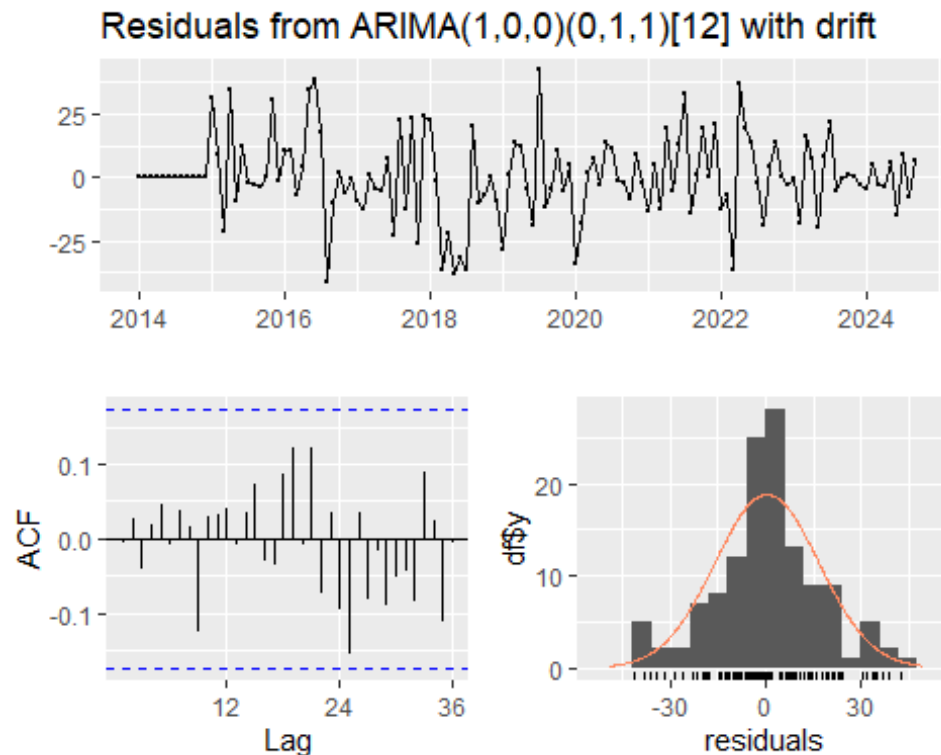
## ARIMA Forecast

**Forecasts from ARIMA(1,0,0)(0,1,1)[12] with drift**



The CI's follow the data and allow for us to see the range in which our data is likely to fall as well as the predicted value that is forecasted. The ARIMA model produced forecast accuracy metrics comparable to Holt-Winters while also generating residuals that behaved similarly to white noise, indicating that the model adequately captured the underlying structure of the data.

## Model Diagnostics By Using Checkresiduals()



```
##  
##  Ljung-Box test  
##  
## data:  Residuals from ARIMA(1,0,0)(0,1,1)[12] with drift  
## Q* = 13.178, df = 22, p-value = 0.9281  
##  
## Model df: 2.    Total lags used: 24
```

Residual diagnostics using the Ljung-Box test indicated no evidence of significant residual autocorrelation. The test produced a p-value of 0.9281, suggesting that the residuals behaved similarly to white noise and that the ARIMA model successfully accounted for the primary patterns within the transaction series.

Additionally, the residual histogram appeared approximately normally distributed, further supporting the adequacy of the forecasting model.

## Conclusion

Overall, the analysis demonstrated that Cincinnati Reds transaction activity follows strong and highly consistent yearly seasonal patterns tied closely to the structure of the MLB calendar. Forecasting models capable of incorporating both trend and seasonality, particularly Holt-Winters and Seasonal ARIMA, substantially outperformed simpler smoothing approaches.

Among the evaluated models, the Seasonal ARIMA framework provided the strongest overall balance between forecast accuracy, seasonal representation, and residual behavior. These findings highlight the importance of seasonality within baseball operations forecasting and demonstrate how time series methods can be used to model organizational transaction behavior over time.